Dual Space.

Let V be a Vector space over a field F.

Def. A linear map $f: V \to F$ is called a linear functional on V.

Example 1. Let $F[x]$ be the set of polynomials over F. For each $\alpha \in F$, define

$$L_\alpha : F[x] \to F ; p(x) \mapsto p(\alpha)$$

is called a 'evaluation at α. Clearly, it is linear functional,

$$L(c \cdot p(x) + q(x)) = c \cdot p(\alpha) + q(\alpha) = c \cdot L(p(x)) + L(q(x)).$$

Example 2. Let $C([a,b])$ be a space of continuous real-valued maps on $[a,b]$. Define

$$L : C([a,b]) \to \mathbb{R} ; f \mapsto \int_a^b f$$

is clearly linear functional by the linearity of integral.

Def. A collection of all linear functional $L(V,F)$ is called the dual space of V, denote as V^*.

Thm. Let V be a finite-dimensional Vector space over F, and let $\beta = \{\alpha_1, \cdots, \alpha_n\}$ be a basis for V.
Then, there exists a unique dual basis $\beta^* = \{f_1, \cdots, f_n\}$ for V^* s.t. $f_i(\alpha_j) = \delta_{ij}$.
For each linear functional $f \in V^*$,

$$f = \sum_{i=1}^{n} f(\alpha_i) \cdot f_i$$

and for each vector $\alpha \in V$,

$$\alpha = \sum_{i=1}^{n} f_i(\alpha) \cdot \alpha_i$$

Thm. Let V be a finite-dimensional Vector space over F, and let $\beta = \{\alpha_1, \dots, \alpha_n\}$ be a basis for V.
Then, there exists a unique dual basis $\beta^* = \{f_1, \dots, f_n\}$ for V^* s.t. $f_i(\alpha_j) = \delta_{ij}$.
For each linear functional $f \in V^*$,

$$f = \sum_{i=1}^{n} f(\alpha_i) \cdot f_i$$

and for each vector $\alpha \in V$,

$$\alpha = \sum_{i=1}^{n} f_i(\alpha) \cdot \alpha_i$$

Prof. By the property of basis, for any $\alpha \in V$, there exist unique scalars $a_1, \dots, a_n \in F$ s.t.

$$\alpha = a_1\alpha_1 + \dots + a_n\alpha_n$$

For each $1 \le i \le n$, define

$$f_i : V \to F : \alpha \mapsto a_i. \quad (\text{or } f_i(\alpha) = ([\alpha]_\beta)_i.)$$

Now, $f_i(\alpha_j) = \delta_{ij}$. And, since $\dim V = \dim V^*$, enough to show that the set

$$\beta^* = \{f_1, \dots, f_n\}$$

is linearly independent. Suppose that for some $c_i \in F$, $\sum_{i=1}^{n} c_i f_i = 0$.
Then, for each $1 \le j \le n$, $\sum_{i=1}^{n} c_i f_i(\alpha_j) = c_j = 0$. Thus proved.
The second assertion can be proved by: if $\alpha \in V$ is represented as

$$\alpha = \sum_{i=1}^{n} x_i \alpha_i \quad (x_i \in F)$$

then, for each $1 \le j \le n$,

$$f_j(\alpha) = f_j\left(\sum_{i=1}^{n} x_i \alpha_i\right) = \sum_{i=1}^{n} x_i f_j(\alpha_i) = x_j$$

thus proved. ☒